

2024



Ginkgo Makes Biology Easier to Engineer and Scales Cost Efficiently Using AWS Batch

Learn how synthetic biology company Ginkgo Bioworks saved up to 90 percent on Amazon EBS storage.

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Up to 90%

reduction in Amazon EBS costs

Up to 60%

reduction in Amazon S3 costs

10x speed improvement

for experiments

Scaled

Met



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Overview

Biotechnology innovator [Ginkgo Bioworks](#)

customer demand in industries such as agriculture, pharmaceuticals, and food. The company, which analyzes and programs organisms, wanted to exponentially increase the amount of scientific engineering it could do. Yet, with cell analytics workloads requiring as much as 1 PB of memory, Ginkgo knew it needed unlimited computing power to drive growth.



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In 2019, Ginkgo migrated its offering to Amazon Web Services (AWS) to unlock cost-effective compute at scale. The biology company now uses a host of AWS services that empower its scientists to focus on innovation. “Using AWS, we achieved a step change in how we compute and scale our biological solutions,” says Dannerys Duran, head of digital operations at Ginkgo. “It improved how we work in labs, how we analyze data, and how we run software. We have genome assembly experiments that went from taking 40 hours to being finished in 4 hours and RNA sequencing datasets where the processing time was reduced from 24 to just a few hours.”



Opportunity | Scaling Synthetic Biology Cost Efficiently Using AWS Batch for Ginkgo

Ginkgo was founded by five scientists from the Massachusetts Institute of Technology (MIT) in 2008, and it uses synthetic biology to provide biological solutions for companies across a wide variety of sectors. For example, Ginkgo’s customers apply its solutions to further drug development or reduce the need for fertilizers in agriculture. Initially, Ginkgo relied on on-premises data centers to power its solutions. However, as the company grew, it was clear that a new approach to high performance computing (HPC) was needed. Because so much of Ginkgo’s work involves biological research and development, the company’s need for computing power to deliver high-quality biology services is highly variable. “Cells just divide. That’s the original exponential growth,” says Chris Mitchell, principal software engineer at Ginkgo. “If you’re dealing with biology, you need to accommodate scale.”

To unlock scalability while freeing up time, Ginkgo migrated its systems to [AWS Batch](#), a service to efficiently run hundreds of thousands of batch computing jobs. “One of the main drivers behind our cloud migration was that we could not handle the data load locally and do all the analysis that our customers need without spending significant resources and time upfront,” says Duran. To get the most out of its services, Ginkgo communicated with AWS to discuss architecture, refine solutions, and optimize spending. Now, Ginkgo uses AWS services such as AWS Batch to take care of the undifferentiated heavy lifting so that Ginkgo teams can focus on delivering high-quality synthetic research and engineering at scale.



Solution | Using AWS Batch to Run Hundreds of Thousands of Jobs

Along with AWS Batch, Ginkgo uses [Amazon Elastic Block Store](#) (Amazon EBS), a scalable block storage service, to optimize its storage volumes. The company runs immense HPC workloads to discover the best way to program organisms, and it uses Amazon EBS to store the petabytes of data it needs for analytics. “This year, we’ve saved up to 90 percent of our Amazon EBS costs by reducing the size of the volumes we use,” says Dima Kovalenko, manager of cloud infrastructure at Ginkgo.

In addition to AWS Batch and Amazon EBS, the company uses several other services to achieve flexibility. For example, it uses [Amazon Simple Storage Service](#) (Amazon S3), a highly secure and available object storage service, to store data cost efficiently after experiments have run their course. More specifically, Ginkgo uses [Amazon S3 Intelligent-Tiering](#) (S3 Intelligent-Tiering), a service that automates storage cost savings by moving data when access patterns change. “Using S3 Intelligent-Tiering, we’ve saved up to 60 percent of our storage costs,” says Kovalenko.

With its solutions running on AWS, Ginkgo’s teams can now focus more closely on science. Rather than dealing with infrastructure management, the company can turn around projects and provide its customers with answers as quickly as possible. Ginkgo has developed an efficient platform that automates much of the lab work that biotechnology companies would otherwise have to do for synthetic biology. “When people come to us with an exciting idea, we empower them to do rapid experimentation to see if their idea is worth pursuing further,” says Kovalenko. “We don’t have to wait 6 months to set up a data center, and that makes a huge impact for our customers.” Using AWS, the company has met its goal of scaling to meet growing customer demand.

Now that its systems are on AWS, Ginkgo can scale to onboard many more customers on its biology platform. “Since going public, we have to comply with Sarbanes-Oxley regulations, and many of our customers have specific security requirements for their programs,” says Duran. “Without migrating to AWS, we couldn’t have accommodated the needs of our internal teams nor those of our customers cost efficiently.”

Outcome | Unlocking Precision Using Amazon EFS

Ginkgo plans to continue developing its systems on AWS. To achieve a greater level of insight into its file usage and optimize even further, the company is migrating to [Amazon Elastic File System](#) (Amazon EFS), an elastic file system. “We’re migrating as many of our active projects as possible to Amazon EFS for the future,” says Kovalenko.



By taking care of the heavy lifting involved in synthetic biology, Ginkgo is positioning itself as a catalyst for innovation across multiple industries. “Imagine if there were an AWS for biology. That’s Ginkgo,” says Sudeep Agarwala, program director at Ginkgo. “We deal with the hardware, technical details, and experiments so that our customers can focus on products.”

About Ginkgo Bioworks

Ginkgo Bioworks, which was founded by five MIT scientists in 2008, is a full-stack synthetic biology company that provides research and development, design, manufacturing, and process optimization services to customers in multiple industries.

AWS Services Used

AWS Batch

AWS Batch lets developers, scientists, and engineers efficiently run hundreds of thousands of batch and ML computing jobs while optimizing compute resources, so you can focus on analyzing results and solving problems.

[Learn more »](#)

Amazon Elastic File System

Amazon Elastic File System (EFS) automatically grows and shrinks as you add and remove files with no need for management or provisioning.

[Learn more »](#)

Amazon S3 Intelligent-Tiering

Amazon S3 Intelligent-Tiering is the only cloud storage class that delivers automatic storage cost savings when data access patterns change, without performance impact or operational overhead.

[Learn more »](#)



Amazon Elastic Block Store

Amazon Elastic Block Store (Amazon EBS) is an easy-to-use, scalable, high-performance block-storage service designed for Amazon Elastic Compute Cloud (Amazon EC2).

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